

Assignment 5

Kumar Kanishk Singh

2024-04-15

Question2

```
#Question 2  
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.2.3
```

```
data("galaxies")  
Y = galaxies  
hist(Y, breaks = 25)  
n = length(Y)
```

```
data= list(Y = Y, N=n, K = 3, alpha = rep(1, 3))
```

```
library(rjags)
```

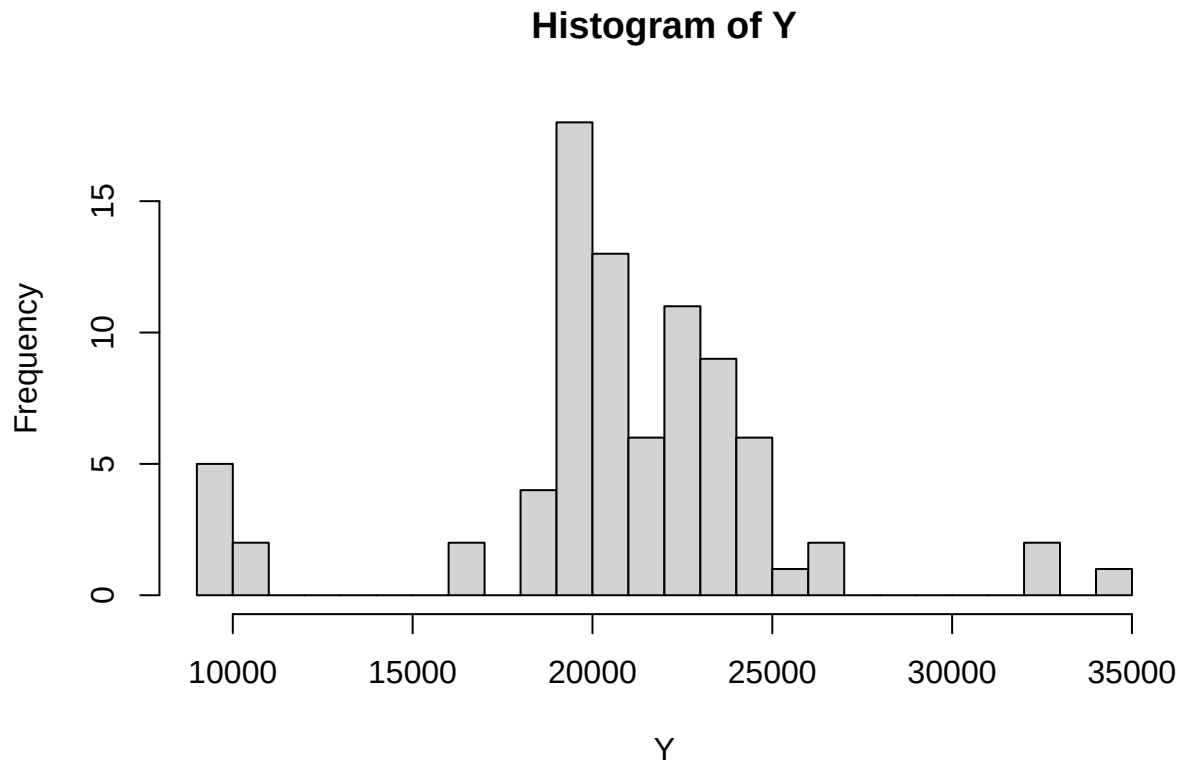
```
## Warning: package 'rjags' was built under R version 4.2.3
```

```
## Loading required package: coda
```

```
## Warning: package 'coda' was built under R version 4.2.3
```

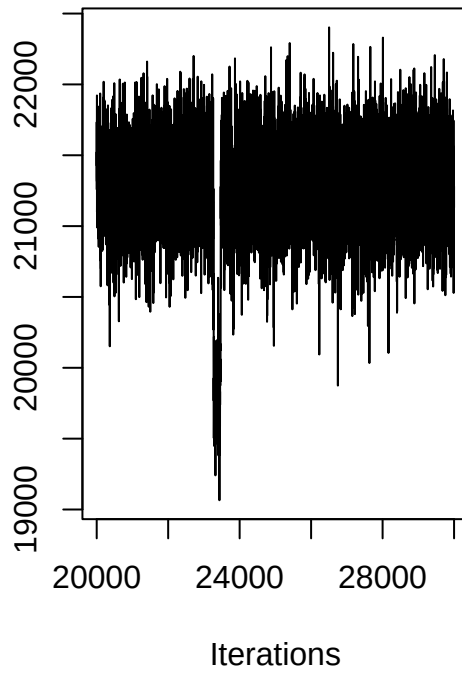
```
## Linked to JAGS 4.3.1
```

```
## Loaded modules: basemod,bugs
```

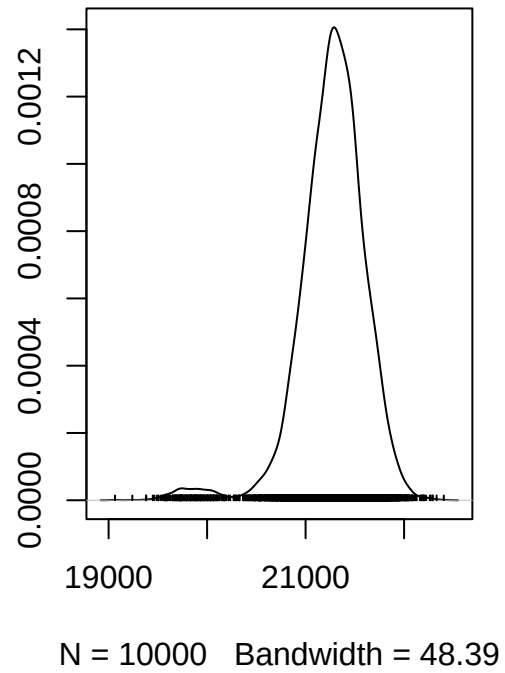


```
model_string = textConnection("model{  
  
# Likelihood  
for (i in 1:N) {  
  Y[i] ~ dnorm(mu[Z[i]], tau[Z[i]])  
  
  Z[i] ~ dcat(theta[])  
}  
  
for (j in 1:K) {  
  mu[j]~ dnorm(0, 1e-8)  
  tau[j]~ dgamma (0.01, 0.01)  
}  
theta[1:K] ~ ddirch(alpha[])  
}")  
  
params = c('mu', 'tau', 'theta')  
  
model = jags.model(model_string, data =data, quiet = TRUE)  
update(model, 2e4)  
  
samples = coda.samples (model, variable.names = params, n.iter = 1e4)  
plot(samples[[1]][,1])
```

Trace of var1

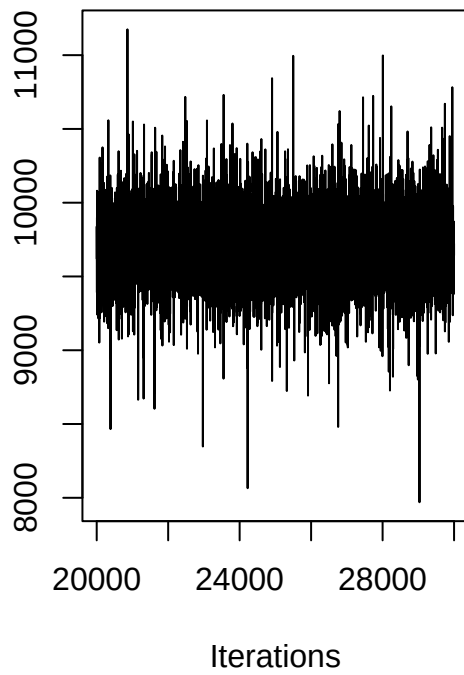


Density of var1

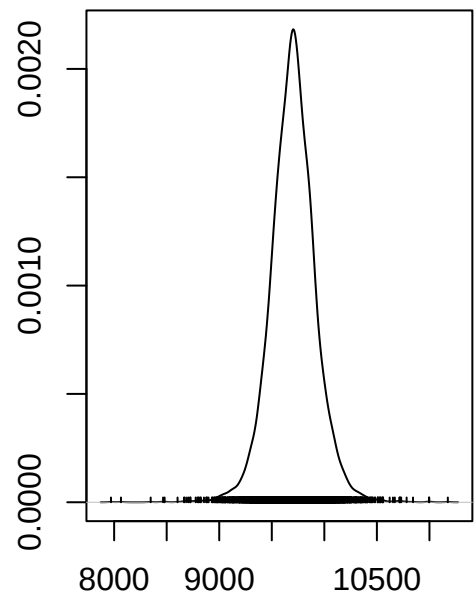


```
plot(samples[[1]][,2])
```

Trace of var1



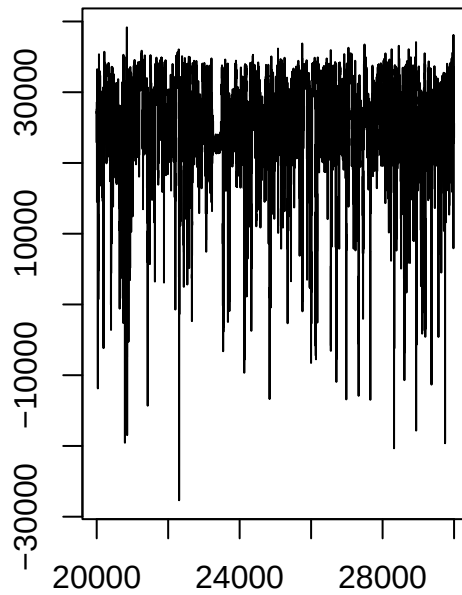
Density of var1



N = 10000 Bandwidth = 32.45

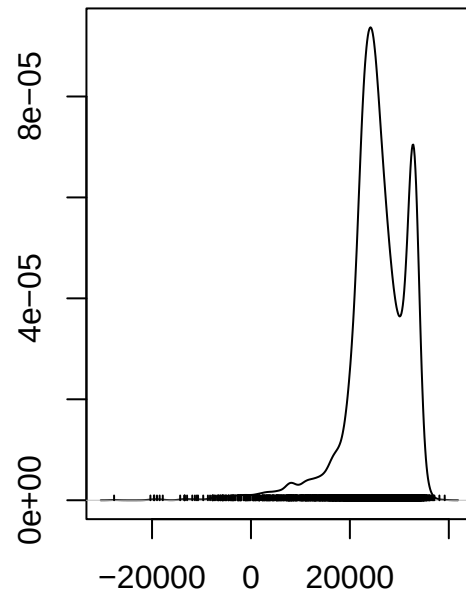
```
plot(samples[[1]][,3])
```

Trace of var1



Iterations

Density of var1



N = 10000 Bandwidth = 900.7

```
y = seq(5e3, 4e4, 100)

mu.post = samples[[1]][,1:3]
tau.post = samples[[1]][,4:6]
theta.post = samples[[1]][,7:9]

S = 1e4

post_density = matrix(NA, nrow = S, ncol = 351)

for(i in 1:S) {
  mu = as.numeric(mu.post[i,])
  sigma = as.numeric(1/ sqrt(tau.post[i,]))
  theta = as.numeric(theta.post[i, ])

  mix_gauss = function(x) {

    theta [1] * dnorm(x, mean = mu[1], sd =sigma[1])
    +
    theta [2] * dnorm(x, mean = mu[2], sd = sigma[2]) +
    theta[3] * dnorm(x, mean = mu[3], sd = sigma[3]) }

  post_density[i,] = mix_gauss(y)
```

```

post_density[i, ] <- sapply(y, mix_gauss)

  #print(paste0('Done:', i))
}

post_median <- apply(post_density, 2, median)
post_2.5.quantile <- apply(post_density, 2, quantile, probs = 0.025)
post_97.5.quantile <- apply(post_density, 2, quantile, probs = 0.975)

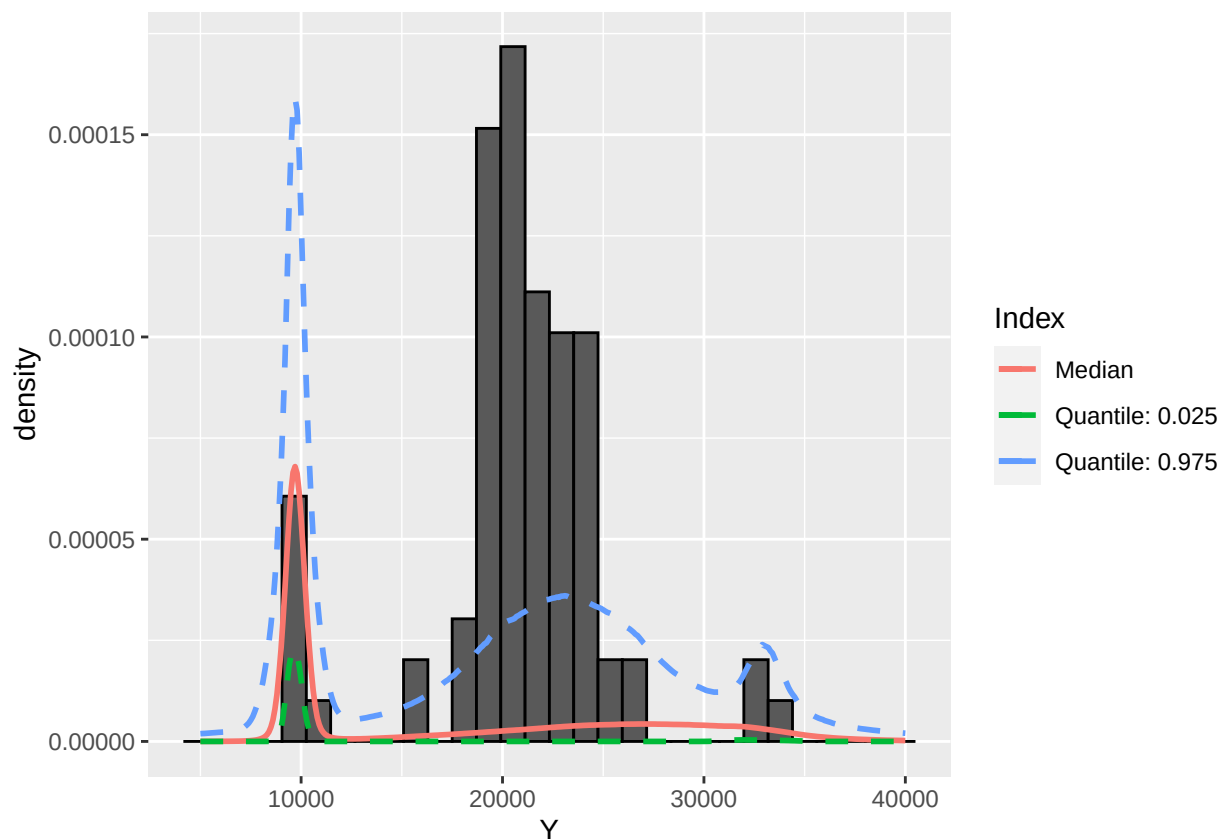
par(mfrow = c(1,1))

library(ggplot2)

ggplot()+
  geom_histogram(aes(x = Y, y =after_stat(density)), col = 'black')+
  geom_line(aes(y, post_median, col = 'Median'), size = 1)+
  geom_line(aes(y, post_2.5.quantile, col = 'Quantile: 0.025'), linetype = 'dashed', size = 1)+
  geom_line(aes(y, post_97.5.quantile, col = 'Quantile: 0.975'), linetype = 'dashed', size = 1)+
  labs(col = 'Index')

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.

```



Question 3

```
set.seed(100)

Y1<-563
Y2<-10

N1<-2820
N2<-27

Y<-c(Y1,Y2)
N<-c(N1,N2)

library(rjags)

#M1
model_string1<-"model{
  #Likelihood
  for(i in 1:length(N))
  {
    Y[i]~dpois(N[i]*lambda[i])
  }

  #Priors
  c<-1
  lambda1~dunif(0,c)
  lambda2~dunif(0,c)

  lambda<-c(lambda1, lambda2)
}"

#M2
model_string2<-"model{
  #Likelihood
  for(i in 1:length(N))
  {
    Y[i]~dpois(N[i]*lambda0)
  }

  #Priors
  c<-1
  lambda0~dunif(0,c)
  lambda<-lambda0
}"

data<-list(Y=Y,N=N)
model1<-jags.model(textConnection(model_string1),
                   data=data,n.chains=1,quiet=TRUE)
update(model1,10000,progress.bar = "none")
sample1<-coda.samples(model1,variable.names=c("lambda"),
                      n.iter=20000,thin=5,progress.bar="none")
lambda.m1<-sample1[[1]]

model2<-jags.model(textConnection(model_string1),
```

```

        data=data,n.chains=1,quiet=TRUE)
update(model1,10000,progress.bar ="none")
sample2<-coda.samples(model2,variable.names=c("lambda"),
        n.iter=20000,thin=5,progress.bar="none")

lambda.m2<-sample2[[1]]

#after thinning,4Kpost-burn-insamplesleft
loglike.m1<-sapply(1:4000,function(iter){
  sum(dpois(Y1,lambda=lambda.m1[iter,1],log=T)+dpois(Y2,lambda.m1[iter,2],log=T))})
loglike.m2<-sapply(1:4000,function(iter){
  sum(dpois(Y1,lambda=lambda.m2[iter,1],log=T)+dpois(Y2,lambda.m2[iter,1],log=T))})

deviance.m1<--2*loglike.m1
deviance.m2<--2*loglike.m2
#DIC
Dbar.m1<-mean(deviance.m1)
Dbar.m2<-mean(deviance.m2)

D.thetahat.m1<-sum(dpois(Y1,lambda=mean(lambda.m1[,1]),log=T)+dpois(Y2,lambda=mean(lambda.m1[,2]),log=T))
D.thetahat.m2<-sum(dpois(Y1,lambda=mean(lambda.m2[,1]),log=T)+dpois(Y2,lambda=mean(lambda.m2[,1]),log=T))

pD.m1<-Dbar.m1-D.thetahat.m1
pD.m2<-Dbar.m2-D.thetahat.m2
DIC.m1<-pD.m1+Dbar.m1
DIC.m2<-pD.m2+Dbar.m2
print(DIC.m1)

## [1] 19690.39

print(DIC.m2)

## [1] 19725.02

#WAIC
loglike1.m1<-sapply(1:4000,function(iter){
  dpois(Y1,lambda=lambda.m1[iter,1],log=T)})
loglike2.m1<-sapply(1:4000,function(iter){
  dpois(Y2,lambda=lambda.m1[iter,2],log=T)})
loglike1.m2<-sapply(1:4000,function(iter){
  dpois(Y1,lambda=lambda.m2[iter,1],log=T)})
loglike2.m2<-sapply(1:4000,function(iter){
  +dpois(Y2,lambda.m2[iter,1],log=T)})

posmeans.m1<-c(mean(loglike1.m1),mean(loglike2.m1))
posmeans.m2<-c(mean(loglike1.m2),mean(loglike2.m2))
posvars.m1<-c(var(loglike1.m1),var(loglike2.m1))
posvars.m2<-c(var(loglike1.m2),var(loglike2.m2))
pW.m1<-sum(posvars.m1)
pW.m2<-sum(posvars.m2)
sum.means.m1<-sum(posmeans.m1)
sum.means.m2<-sum(posmeans.m2)
WAIC.m1<--2*sum.means.m1+2 * pW.m1
WAIC.m2<--2*sum.means.m2+2 * pW.m2

```

```

print(WAIC.m1)

## [1] 9027.741

print(WAIC.m2)

## [1] 9029.394

OUT<-rbind(c(DIC.m1,WAIC.m1), c(DIC.m2,WAIC.m2))
OUT<-round(OUT,2)
rownames(OUT)<-c("Model1","Model2")
colnames(OUT)<-c("DIC","WAIC")
library(kableExtra)

## Warning: package 'kableExtra' was built under R version 4.2.3

kable(OUT)

```

	DIC	WAIC
Model1	19690.39	9027.74
Model2	19725.02	9029.39

We can see that model 1 has lower DIC and WAIC than model 2. Hence, model 1 is preferred.